

BEIXUAN DONG

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EDUCATION

University of Alberta, Edmonton, Canada	Sep 2023–Present
Ph.D. in Engineering Management, Department of Mechanical Engineering GPA: 3.8/4.0	
Supervisors: Dr. Xinming Li and Dr. Lingzi Wu	
Central South University, Changsha, China	Sep 2020–Jun 2023
M.Sc. in Engineering Management GPA: 3.6/4.0	
Supervisor: Dr. Ming Shan	
Central South University of Forestry and Technology (CSUFT), Changsha, China	Sep 2016–Jun 2020
B.Eng. in Engineering Management GPA: 4.0/4.0 Rank: 1/73	

VISITING RESEARCH APPOINTMENTS

University of Washington, Seattle, USA	Jun–Aug 2025
Visiting Ph.D. Student, Department of Construction Management Host: Dr. Lingzi Wu	
National University of Singapore, Singapore	Dec 2021–Jul 2022
Visiting Master's Student, Department of the Built Environment Host: Dr. Bon-Gang Hwang	

RESEARCH INTEREST

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- AI application for infrastructure disaster management
 - Infrastructure resilience
 - Data-driven modeling

PUBLICATIONS

Peer-reviewed-journal papers

- [1]. Dong, B., Wu, L. & Li, X. (2026). Towards equitable post-disaster highway restoration: A GNN-based framework for segment-level prioritization. *Transportation Research Part D: Transport and Environment*. (Revision, Aug 2027, IF: 8.7)
- [2]. Dong, B., Li, X., & Wu, L. (2026). A GNN-Based Framework for Assessing Flood Impacts on Highway Networks: Integrating Network Structural, Functional, and Social Features. *Journal of Management in Engineering*, 42(4), 04026010. <https://doi.org/10.1061/JMENEA.MEENG-7198> (IF: 6.0)
- [3]. Dong, B., Ding, S., Wu, L., & Li, X. (2025). Short-term natural disaster impacts on transportation

infrastructure: a systematic review. *Natural Hazards*, 121(15), 17321-17362. <https://doi.org/10.1007/s11069-025-07504-9> (IF: 4.5)

- [4]. **Dong, B. X.**, Shan, M., Cui, J. H., & Hwang, B. G. (2025). Examining mental resilience of contractor in urban rail transit construction project under disruptive events. *Tunnelling and Underground Space Technology*, 163, 106725. <https://doi.org/10.1016/j.tust.2025.106725> (IF: 8.7)
- [5]. **Dong, B. X.**, Shan, M., & Hwang, B. G. (2022). Simulation of transportation infrastructures resilience: a comprehensive review. *Environmental Science and Pollution Research*, 29(9), 12965-12983. <https://doi.org/10.1007/s11356-021-18033-w>

Peer-Reviewed Conference Proceedings

- [1]. **Dong, B.**, Wu, L., & Li, X. Flood Impact Assessment on Highway Networks Considering Socioeconomic Vulnerability: Evidence from Harris County. 43rd International Symposium on Automation and Robotics in Construction (ISARC), June 22-26, 2026, Singapore. <https://doi.org/10.22260/ISARC2026/0191>
- [2]. **Dong, B.**, Zhou, C., Gue, B., Li, X., & Wu, L. Impacts Identification of Supplier Reliability on Project Duration in Heavy Industrial Construction Supply Chain Using Discrete Event Simulation. The 10th Joint CRC/CSCE Construction Speciality Conference, 28-31st July, 2025, Montreal, Quebec, Canada. <https://doi.org/10.22260/CRC-CSCE-2025/0081>
- [3]. **Dong, B.**, Wu, L., & Li, X. Short-Term Disaster Impacts on Transportation Infrastructure: A Review of Emerging Technologies and Approaches. *Computing in Civil Engineering*, July 28–31, 2024, Pittsburgh, Pennsylvania, USA. <https://doi.org/10.1061/9780784486139.031>

Poster presentation

- [1]. **Dong, B.**, Zhou, C., Gue, B., Li, X., & Wu, L. Impacts Identification of Supplier Reliability on Project Duration in Heavy Industrial Construction Supply Chain Using Discrete Event Simulation [Poster presentation], Construction Innovation Center (CIC), June 12-13, 2024, Edmonton, Canada. <https://constructioncic.ca/forum2024/>

Dissertation

- [1]. **Dong, B.X.** (2023) Multi-agent-based Simulation of Resilience in Urban Rail Transit Construction Projects, M.S. Dissertation, Central South University, Changsha, China

Book & Book Chapter

- [1]. Yu D.M., Shan M., Li Y.G. (2021). *International Railway Project Risk Management*. Central South University Press.

Note: As a helper to draft Chapter 12 of the book, which is Construction Insurance and Surety Bonds.

RESEARCH EXPERIENCES

Data-Driven Modeling for Disaster Resilience in Road Networks

2023-present

Advised by Dr. Xinming Li and Dr. Lingzi Wu

- Synthesized methods for assessing short-term natural-disaster impacts on transportation infrastructure through a systematic literature review.
- Developed a graph neural network framework integrating structural, functional, and social-vulnerability features to assess and predict flood impacts on highway networks.
- Designed an equity-aware framework for prioritizing road restoration based on roadway function and the accessibility needs of socially vulnerable communities.
- Proposed an accessibility-based post-disaster road restoration approach, where each decision aims to maximize the marginal accessibility gain while accounting for dynamic social vulnerability.

Resilience of Urban Rail Transit Construction Projects

2020-2023

Advised by Dr. Ming Shan and Dr. Bon-Gang Hwang

- Identified attributes shaping the physical resilience of urban rail transit projects and the mental resilience of project contractors.
- Developed an agent-based model in AnyLogic to simulate interactions among disruptive events, infrastructure performance, and stakeholder well-being.
- Applied the model to a real-world case and derived strategies for improving project resilience.

GRANT PROPOSAL PARTICIPATION

Resilience in Urban Rail Transit Construction Projects, granted by the Natural Science Foundation of Hunan Province. PI: Dr. Ming Shan, CNY 150000

2022

TEACHING EXPERIENCE

Teaching Assistant, ENG M 632: Project Risk Management, University of Alberta

Jan–Apr 2026

- Graded assignments and provided feedback

Teaching Assistant, ENG M 620: Engineering Economic Analysis, University of Alberta

Jan–Apr 2026

- Graded assignments and provided feedback

Teaching Assistant, ENG M 530: Engineering Project Management, University of Alberta

Sep–Dec 2025

- Assisted in developing and revising assignments
- Supported students and graded assignments

Teaching Assistant, ENG M 530: Engineering Project Management, University of Alberta

May–Jul 2024

- Independently delivered instruction on project scheduling
- Supported students and graded assignments

AWARDS AND SCHOLARSHIPS

Alberta Innovates Graduate Scholarship, University of Alberta

2025–2027 | CAD 24,000

Academic Travel Grant, University of Alberta

2026 | CAD 500

Academic Travel Grant, University of Alberta

2025 | CAD 500

Second-Class Merit-Based Scholarship, Central South University

2021, 2022

First-Class Merit-Based Scholarship, Central South University

2020

Outstanding Undergraduate Graduate of Hunan Province

2020 | Top 5%

National Scholarship for Undergraduate Students, China

2019 | CNY 8,000 | Top 1%

First-Class Merit-Based Scholarship, CSUFT

2016–2018 | Top 5%

PROFESSIONAL SERVICE

Professional affiliations

Student Member – American Society of Civil Engineers (ASCE)

Reviewer Services

ASCE Journal of Construction Engineering and Management	2024–Present
Elsevier Advanced Engineering Informatics	2026–Present
Elsevier Cities	2026–Present
Elsevier Cleaner Cities	2026–Present
ASCE Journal of Computing in Civil Engineering	2026–Present
43rd International Symposium on Automation and Robotics in Construction	2026
CI & CRC Joint Conference, San Antonio, TX, USA	2026
42nd International Symposium on Automation and Robotics in Construction	2025
41st International Symposium on Automation and Robotics in Construction	2024

TECHNICAL SKILLS AND LANGUAGES

Programming: Python, Java, C++, MATLAB

Modeling and Methods: graph neural networks, deep learning, geospatial network analysis, agent-based simulation, discrete-event simulation

Software: Codex, AnyLogic, Autodesk Revit, AutoCAD

Languages: Chinese (native); English (professional working proficiency)